As you can see, the process starts by fetching previous conversation messages from a database or any storage solution you are using. This message retrieval will provide the context needed for the current interaction. Messages can be stored using a unique identifier for each session, and each user might have one or more sessions stored (like how ChatGPT and other services offer a way to select previous chat histories).

Once you have the past messages, convert them into instances of `ChatMessageHistory`. This allows you to manage and utilize the conversation history effectively. Next, create a chain out of `ChatMessageHistory`. Use the `ConversationChain` class provided, passing in the language model and the memory instance. This chain will handle interactions while maintaining context from previous messages.

Then you can perform queries based on conversation history. With everything set up, you can invoke the `ConversationChain` to handle user queries, using the stored conversation history for context. You can also include the chain in the context of other prompts to improve the accuracy of the results. These steps can be executed with just a few lines of code, as shown in the following listing.

Listing 6.8 Performing queries with a conversation memory

Imports message classes for system, human, and AI interactions

```
import { SystemMessage, HumanMessage, AIMessage } from "@langchain/core/
messages";
import { ChatMessageHistory } from "langchain/stores/message/in_memory";

const llm = new ChatGoogleGenerativeAI({ apiKey });
const pastMessages = [];
pastMessages.push(new SystemMessage("You are a helpful assistant.
Answer all questions to the best of your ability."));
pastMessages.push(new HumanMessage("Hi! I'm Jim."));

const memory = new BufferMemory({
    chatHistory: new ChatMessageHistory(pastMessages),
  });

const res1 = await chain.invoke({
input: "Can you give me an example of AI?" });
console.log({ res1 });
  /*
  Expected Output:
  {
    res1: {
      text: "Sure! An example of AI is a virtual
assistant like Google Assistant."
    }
  }
  */

const res2 = await chain.invoke({ input: "What did I just ask you?" });
console.log({ res2 });
```

Initializes a memory buffer to store chat history

Logs the response from the assistant regarding the previous question